

Our residual is given by

$$R_f = f(Y_{t+\Delta t}) - f(X_t) - \int_t^{t+\Delta t} \mathcal{L}_\theta f(X_s) ds \quad (1)$$

so we find that

$$\mathbb{E}[R_f] = \mathbb{E}_{p_{t+\Delta t}}[f] - \mathbb{E}_{p_t}[f] - \Delta t \mathbb{E}_{p_t}[\mathcal{L}_\theta f] \quad (2)$$

and so if $\mathbb{E}[R_f]$ is minimized, then

$$\frac{\mathbb{E}_{p_{t+\Delta t}}[f] - \mathbb{E}_{p_t}[f]}{\Delta t} = \mathbb{E}_{p_t}[\mathcal{L}_\theta f] \quad (3)$$

But the Fokker-Planck equation now gives us that

$$\frac{d}{dt} \mathbb{E}[f(X_t)] = \mathbb{E}[\mathcal{L}f(X_t)] \quad (4)$$

so minimizing the residual enforces

$$\mathbb{E}[\mathcal{L}_\theta f] = \mathbb{E}_{p_t}[\mathcal{L}f] \quad (5)$$